IEAP RStudio Series 03

Data Wrangling and ANOVA

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```r
library(tidyverse)
library(ez)
library(rstatix)
```

GitHub repository: https://github.com/jenniferhoumard/IEAP-Series03-RStudio

1 ANOVA: Analysis of Variance

1.1 Think before coding

1.1.1 Scientific question

We want to test whether movement time increases from ID1 to ID2, and whether the size of this increase differs between groups. This corresponds to the GROUP × ID interaction effect.

1.1.2 How to visualize the effect of difficulty?

The ANOVA p-value tells us whether an effect is statistically significant, while the generalized eta-squared (GES) effect size indicates its magnitude. To identify which specific groups differ, post-hoc pairwise comparisons would be needed. These comparisons are a possible follow-up analysis and are not performed here.

1.1.3 How to test whether there is a significant difference?

We use a mixed-design ANOVA to examine the effects of group and difficulty on movement time.

GROUP is the between-subject factor because each participant belongs to one group. ID is the within-subject factor because each participant performs both difficulty conditions included in our cleaned dataset.

The GROUP × ID interaction is especially important. It tests whether the effect of difficulty on movement time differs between the three groups.

1.1.4 How to report the results?

The article uses a significance threshold of = 0.01. Therefore, p < 0.01 indicates a statistically significant result, whereas p 0.01 indicates that the result is not statistically significant according to this threshold.

We report the ANOVA results using the F-statistic, degrees of freedom, p-value, and generalized eta-squared (GES) effect size for GROUP, ID, and the GROUP × ID interaction.

1.2 What libraries are required to answer the above questions?

The analysis uses the following R packages:

- `tidyverse` is used to import, inspect, reorganize, and visualize the data.
- `rstatix` is used to perform the ANOVA and extract the statistical results.
- `ez` is an additional package available for statistical analysis.

These packages support the data preparation, statistical analysis, and presentation of the results.

1.3 Data reorganization

1.3.1 Data preparation

Before running the statistical analysis, the dataset was checked to make sure the data were complete and organized correctly.

The experiment originally included:

- three groups: `ap`, `cp`, and `pp`
- three ID conditions: `ID1`, `ID2`, and `ID3`
- up to three repetitions: `R1`, `R2`, and `R3`

However, not all participants completed every ID condition and every repetition. Therefore, the dataset had to be checked for missing or incomplete conditions before creating the final dataset used for the analysis.

The groups are:

- `ap` = healthy adult group
- `cp` = healthy young group
- `pp` = stroke group

1.3.1.1 Import and inspect the data

The raw dataset was first imported and inspected to understand its structure, variables, and dimensions.

```
data_raw <- read.table("data/Results.csv", header = TRUE)

head(data_raw) # display data
```

```
  GROUP  SUBJ TRIAL ORI     DIR  ID REP NbVelPeaks MovementTime
1    ap apS01   T01   W outward ID1  R1          2         1.23
2    ap apS01   T01   W  inward ID1  R1          2         1.26
3    ap apS01   T01  NW outward ID1  R1          3         1.14
4    ap apS01   T01  NW  inward ID1  R1          2         0.90
5    ap apS01   T01   N outward ID1  R1          3         1.11
6    ap apS01   T01   N  inward ID1  R1          3         0.81
```

```
str(data_raw) # structure of data
```

```
'data.frame':   7560 obs. of  9 variables:
 $ GROUP       : chr  "ap" "ap" "ap" "ap" ...
 $ SUBJ        : chr  "apS01" "apS01" "apS01" "apS01" ...
 $ TRIAL       : chr  "T01" "T01" "T01" "T01" ...
 $ ORI         : chr  "W" "W" "NW" "NW" ...
 $ DIR         : chr  "outward" "inward" "outward" "inward" ...
 $ ID          : chr  "ID1" "ID1" "ID1" "ID1" ...
 $ REP         : chr  "R1" "R1" "R1" "R1" ...
 $ NbVelPeaks  : num  2 2 3 2 3 3 1 2 4 1 ...
 $ MovementTime: num  1.23 1.26 1.14 0.9 1.11 0.81 1.08 0.81 1.32 1.02 ...
```

```
dim(data_raw) # dimension of data (rows, columns)
```

```
[1] 7560    9
```

1.3.1.2 Check GROUP × ID completeness

Next, the number of observations/rows for each combination of `GROUP` and `ID`:

```
data_raw %>%
  count(GROUP, ID)
```

```
  GROUP  ID    n
1    ap ID1  740
2    ap ID2  740
3    ap ID3  740
4    cp ID1 1140
5    cp ID2 1140
6    cp ID3 1140
7    pp ID1  760
8    pp ID2  760
9    pp ID3  400
```

For the `ap` group, `ID1`, `ID2`, and `ID3` each contain 740 observations. For the `cp` group, all three IDs each contain 1140 observations. This shows that the three ID conditions are complete within these two groups.

For the stroke group (`pp`), `ID1` and `ID2` each contain 760 observations, while `ID3` contains only 400 observations. This shows that `ID3` is incomplete for the `pp` group because some stroke participants did not complete this condition.

Therefore, `ID3` will be removed before the ANOVA analysis so that the same ID conditions are available across all three groups.

1.3.1.3 Remove the incomplete ID3 condition

The previous check showed that `ID3` was incomplete for the stroke group (`pp`). Therefore, all observations belonging to `ID3` were removed before continuing the analysis.

```r
data_clean <- data_raw %>%
  filter(ID != "ID3")

head(data_clean)
```

```
  GROUP  SUBJ TRIAL ORI     DIR  ID REP NbVelPeaks MovementTime
1    ap apS01   T01   W outward ID1  R1          2         1.23
2    ap apS01   T01   W  inward ID1  R1          2         1.26
3    ap apS01   T01  NW outward ID1  R1          3         1.14
4    ap apS01   T01  NW  inward ID1  R1          2         0.90
5    ap apS01   T01   N outward ID1  R1          3         1.11
6    ap apS01   T01   N  inward ID1  R1          3         0.81
```

1.3.1.4 Check repetitions

```
data_clean %>% count(GROUP, REP)
```

```
  GROUP REP   n
1    ap  R1 520
2    ap  R2 520
3    ap  R3 440
4    cp  R1 760
5    cp  R2 760
6    cp  R3 760
7    pp  R1 760
8    pp  R2 760
```

After removing `ID3`, `ID1` and `ID2` contain the same number of observations within each group.

The total number of observations differs between groups because the groups contain different numbers of participants. This does not mean that the ID data are incomplete.

The next step is therefore to check whether the repetitions (`REP`) are complete within each group.

1.3.1.5 Remove the incomplete R3 repetition

The previous check showed that the third repetition (`R3`) was not completed consistently across all groups. Therefore, `R3` was removed from the dataset.

```
data_final <- data_clean %>% filter(REP != "R3")  #remove all rows belonging to R3

head(data_final)
```

```
  GROUP  SUBJ TRIAL ORI     DIR  ID REP NbVelPeaks MovementTime
1    ap apS01   T01   W outward ID1  R1          2         1.23
2    ap apS01   T01   W  inward ID1  R1          2         1.26
3    ap apS01   T01  NW outward ID1  R1          3         1.14
4    ap apS01   T01  NW  inward ID1  R1          2         0.90
5    ap apS01   T01   N outward ID1  R1          3         1.11
6    ap apS01   T01   N  inward ID1  R1          3         0.81
```

The number of observations was then checked again for each group and repetition:

```r
data_final %>% count(GROUP, REP)
```

```
  GROUP REP   n
1    ap  R1 520
2    ap  R2 520
3    cp  R1 760
4    cp  R2 760
5    pp  R1 760
6    pp  R2 760
```

1.3.1.6 Check participants by GROUP

Finally, the number of participants in each group was checked.

```r
data_final %>% distinct(GROUP, SUBJ) %>% count(GROUP)
```

```
  GROUP  n
1    ap 13
2    cp 19
3    pp 19
```

1.3.1.7 Create the cleaned dataset

The final cleaned dataset is `data_final`.

It contains:

- all three groups: `ap`, `cp`, and `pp`
- two complete ID conditions: `ID1` and `ID2`
- two complete repetitions: `R1` and `R2`

`ID3` was removed because it was incomplete for the stroke group (`pp`), and `R3` was removed because it was not completed consistently across all groups.

The cleaned dataset is now ready for the statistical analysis.

1.3.2 ANOVA analysis

Now that we have explored and filtered the data, we can proceed with the ANOVA analysis.

1.3.2.1 Prepare the data for ANOVA

To avoid giving excessive weight to repeated trials (at least 2 for each direction as indicated in the experimental design), movement time was averaged for each participant (SUBJ), group (GROUP) and index of difficulty (ID). The resulting dataset contained one mean movement time value per participant and difficulty level and was used for the subsequent statistical analysis.

1.3.2.2 Run the ANOVA

```
S <- data_final|>
  group_by(GROUP, SUBJ, ID) |>
  summarise(MT = mean(MovementTime), .groups = "drop")
res <- anova_test(data = S, dv = MT, wid = SUBJ,
                  between = GROUP, within = ID)
anova_tab <- get_anova_table(res)

anova_tab
```

```
ANOVA Table (type III tests)

    Effect DFn DFd      F        p p<.05   ges
1    GROUP   2  48 37.244 1.72e-10     * 0.575
2       ID   1  48 26.109 5.54e-06     * 0.066
3 GROUP:ID   2  48 11.581 7.87e-05     * 0.059
```

1.3.2.3 Interpret the ANOVA results

Group effect : The 3 groups (ap, cp, pp) were compared to each other. The ANOVA results showed a significant effect of GROUP on movement time $F(2,48) = 37.24$, $p < 0.001$. This indicates that the mean movement time differed significantly between the three groups.

ID effect : The two levels of ID (ID1, ID2) were compared to each other. The ANOVA results showed a significant effect of ID on movement time ($F(1,48) = 26.11$, $p < 0.001$). This indicates that the mean movement time differed significantly between the two levels of ID. When the ID increased, the movement time increased as well.

Crossed effect (GROUP $\times$ ID) : The interaction between GROUP and ID was the most significant ($F(2,48) = 11.58$, $p < 0.001$). This indicates that the effect of ID on movement time differed between the three groups and was way more important on the stroke group (pp). In other words, the difference in mean movement time between ID1 and ID2 was not the same for all groups.

In conclusion, the effect of difficulty is not the same for all groups, and the effect of group is not the same for all levels of difficulty. This suggests that the relationship between movement time and difficulty level is influenced by the group to which participants belong.

1.3.2.4 Plot the GROUP × ID interaction

```
interaction_plot_Group_ID <- ez::ezPlot(
  data = S,
  dv = .(MT),
  wid = .(SUBJ),
  within = .(ID),
  between = .(GROUP),
  x = .(ID),
  split = .(GROUP)
)
```

```
Warning: Converting "SUBJ" to factor for ANOVA.
```

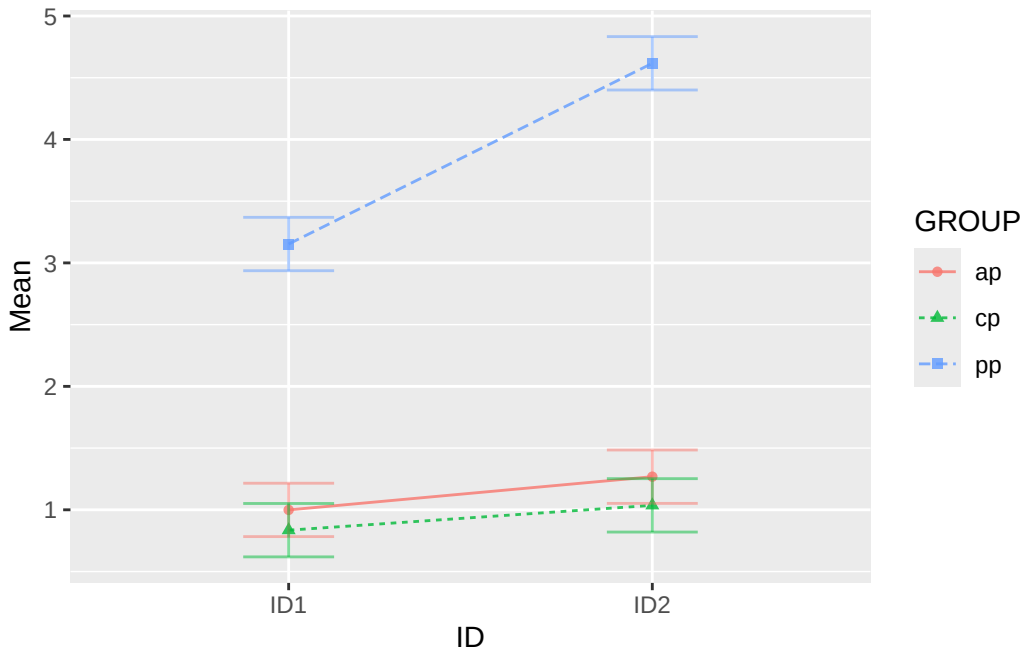
```
Warning: Converting "ID" to factor for ANOVA.
```

```
Warning: Converting "GROUP" to factor for ANOVA.
```

```
Warning: Data is unbalanced (unequal N per group). Make sure you specified a
well-considered value for the type argument to ezANOVA().
```

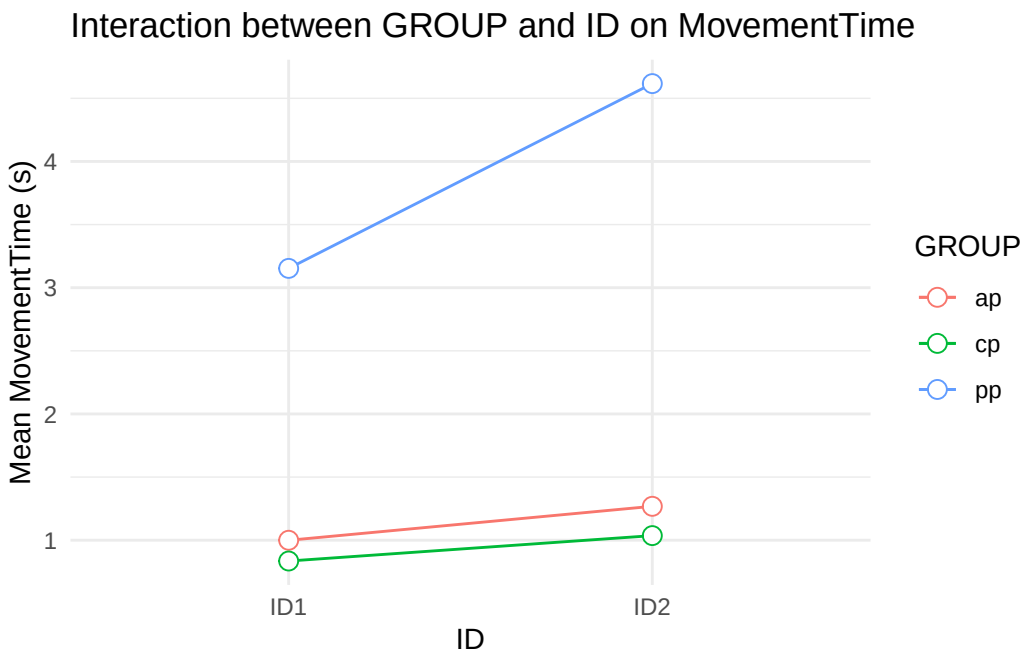
```
Warning in ezStats(data = data, dv = dv, wid = wid, within = within,
within_full = within_full, : Unbalanced groups. Mean N will be used in
computation of FLSD
```

```
interaction_plot_Group_ID
```




```
ggplot(data_final, aes(x = ID, y = MovementTime, color = GROUP)) +
  stat_summary(fun = mean, geom = "line", aes(group = GROUP)) +
  stat_summary(fun.data = mean_sdl, fun.args = list(mult = .5),
               geom = "errorbar", width = 0.01) +
  stat_summary(fun = mean, geom = "point", shape = 21, fill = "white", size = 3) +
  labs(title = "Interaction between GROUP and ID on MovementTime",
       x = "ID", y = "Mean MovementTime (s)") +
  theme_minimal()
```

```
Warning: Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```



Graphically, we can see that a significant GROUP × ID interaction was observed, $F(2,48) = 11.58$, $p < 0.001$. As shown in the graph displayed above, movement time increased with task difficulty in all groups, but the increase was larger for the stroke group (PP) than for the healthy groups (AP and CP).

1.4 Run the linear regression by GROUP

The aim is to see how MovementTime changes when the difficulty (ID) increases, and whether this change is different between the three groups.

A separate linear regression is calculated for each group:

ap = healthy adults

cp = healthy young adults

pp = stroke group

The regression equation is:

$y = mx + b$

$MT = a + b(ID)$

where:

MT = Movement Time

a = intercept

b = slope

ID = Index of Difficulty

The slope tells us how much Movement Time changes when ID increases.

1.4.1 Regression equations

For the ANOVA, `ID` was treated as a categorical variable with two conditions: `ID1` and `ID2`.

For the linear regression, we need to convert the categorical `ID` variable to a numerical variable with the actual Index of Difficulty values:

- `ID1` = 3.80
- `ID2` = 5.43

A new variable called `ID_numeric` is therefore created:

```
regression_data <- S %>%
  mutate(ID_numeric = case_when(ID == "ID1" ~ 3.80, ID == "ID2" ~ 5.43))
# mutate() creates the new column `ID_numeric`.
# case_when()` replaces each ID condition with its corresponding numerical value


# Check that the correct numerical value was assigned to each ID
regression_data %>% distinct(ID, ID_numeric)
```

```
# A tibble: 2 x 2
  ID    ID_numeric
  <chr>      <dbl>
1 ID1          3.8
2 ID2          5.43
```

The original `ID` column is kept, while `ID_numeric` is added for the regression analysis.

A separate linear regression is then calculated for each group:

```r
# lm = linear model, MT = y & ID_numeric = x.


# 1. Healthy adult group
regression_ap <- lm(MT ~ ID_numeric, data = regression_data %>% filter(GROUP == "ap")) # only

# 2. Healthy young group
regression_cp <- lm(MT ~ ID_numeric, data = regression_data %>% filter(GROUP == "cp")) # only

# 3. Stroke group
regression_pp <- lm(MT ~ ID_numeric, data = regression_data %>% filter(GROUP == "pp")) # only


show(regression_ap)
```

```
Call:
lm(formula = MT ~ ID_numeric, data = regression_data %>% filter(GROUP ==
    "ap"))

Coefficients:
(Intercept)   ID_numeric
     0.3719       0.1651
```

```r
show(regression_cp)
```

```
Call:
lm(formula = MT ~ ID_numeric, data = regression_data %>% filter(GROUP ==
    "cp"))

Coefficients:
(Intercept)   ID_numeric
     0.3657       0.1235
```

```r
show(regression_pp)
```

```
Call:
lm(formula = MT ~ ID_numeric, data = regression_data %>% filter(GROUP ==
```

```
    "pp"))

Coefficients:
(Intercept)   ID_numeric
    -0.2592       0.8979
```

For each regression:

- `MT` is the Movement Time
- `ID_numeric` is the numerical Index of Difficulty
- `filter()` selects only one group

Therefore, one regression equation is calculated separately for `ap`, `cp`, and `pp`.

The regression equations are:

Healthy adults (`ap`): $MT = 0.372 + 0.165(ID)$

Healthy young adults (`cp`): $MT = 0.366 + 0.123(ID)$

Stroke group (`pp`): $MT = -0.259 + 0.898(ID)$

All three slopes are positive. This means that Movement Time increases when the Index of Difficulty increases.

The stroke group (`pp`) has the largest slope (`0.898`). This means that Movement Time increases more strongly with difficulty in the stroke group than in the two healthy groups.

1.5 Save a graph in a PDF file for publication

A graph is created to show how mean Movement Time changes with ID for each group.

The graph includes:

- the mean Movement Time for each ID
- one line for each group
- error bars showing the variability
- dashed regression lines
- the regression equation for each group

This allows the measured data and the linear regression results to be compared visually.

1.5.1 Create the publication figure

First, the mean Movement Time and standard deviation are calculated for each `GROUP × ID` combination:

```
publication_data <- regression_data %>% group_by(GROUP, ID_numeric) %>%
  summarise(mean_MT = mean(MT), sd_MT = sd(MT), .groups = "drop")

show(publication_data)
```

```
# A tibble: 6 x 4
  GROUP ID_numeric mean_MT sd_MT
  <chr>      <dbl>   <dbl> <dbl>
1 ap           3.8    0.999 0.201
2 ap           5.43   1.27  0.295
3 cp           3.8    0.835 0.121
4 cp           5.43   1.04  0.108
5 pp           3.8    3.15  1.47
6 pp           5.43   4.62  2.41
```

This creates one row for each group and ID containing:

- `mean_MT` = average Movement Time
- `sd_MT` = standard deviation of Movement Time

`GROUP` is converted to a factor to explicitly tell R that it contains categories. The levels are ordered as `pp`, `ap`, and `cp` so that the groups appear in this same order in the graph and legend:

```
publication_data$GROUP <- factor(publication_data$GROUP, levels = c("pp", "ap", "cp"))
```

The cleaned dataset contains many Movement Time measurements for each group and ID.

For the final figure, these measurements are summarized so that there is one mean value for each `GROUP × ID` combination.

`ID1` and `ID2` are first changed to their numerical values (`3.80` and `5.43`) so that ID can be used as a numerical variable in the regression graph.

For each group and ID, the mean Movement Time and standard deviation are then calculated.

The resulting `interaction_data` table contains the values needed to plot the mean points and the `Mean ± 0.5 × SD` error bars.

```r
interaction_data <- data_final %>%
  mutate(ID_numeric = case_when(ID == "ID1" ~ 3.80, ID == "ID2" ~ 5.43)) %>%
  group_by(GROUP, ID_numeric) %>%
  summarise(mean_MT = mean(MovementTime), sd_MT = sd(MovementTime), .groups = "drop")

show(interaction_data)
```

```
# A tibble: 6 x 4
  GROUP ID_numeric mean_MT sd_MT
  <chr>      <dbl>   <dbl> <dbl>
1 ap          3.8    0.999 0.285
2 ap          5.43   1.27  0.427
3 cp          3.8    0.835 0.460
4 cp          5.43   1.04  0.401
5 pp          3.8    3.15  2.40
6 pp          5.43   4.62  3.51
```

1.6 Export the graph as a PDF file for publication

The publication graph is created using `publication_data`:

```r
# 2. Create the ggplot
publication_plot <- ggplot(interaction_data, aes(x = ID_numeric, y = mean_MT, color = GROUP))

  # 3. Dashed regression lines across the full x-axis
  geom_smooth(aes(group = GROUP), method = "lm", se = FALSE, # don't draw confidence interval
              fullrange = TRUE, linetype = "dashed", linewidth = 0.5) +
              # extend regression line beyond full range

  # 4. Solid line connecting the two measured mean values
  geom_line(aes(group = GROUP), linewidth = 0.6) +

  # 5. Error bars showing mean ± 0.5 SD
  geom_errorbar(aes(ymin = mean_MT - 0.5 * sd_MT,
                    ymax = mean_MT + 0.5 * sd_MT),
                width = 0.08) +

  # Mean values
  geom_point(shape = 21, fill = "white", size = 3, stroke = 1) +

  # Regression equations
```

```
  annotate("text", x = 0.5, y = 5.5, label = "MT = -0.259 + 0.898 * ID",
           color = "red", hjust = 0, size = 4) +
  annotate("text", x = 0.5, y = 4.5, label = "MT = 0.372 + 0.165 * ID",
           color = "green3", hjust = 0, size = 4) +
  annotate("text", x = 0.5, y = 3.5, label = "MT = 0.366 + 0.123 * ID",
           color = "blue", hjust = 0, size = 4) +

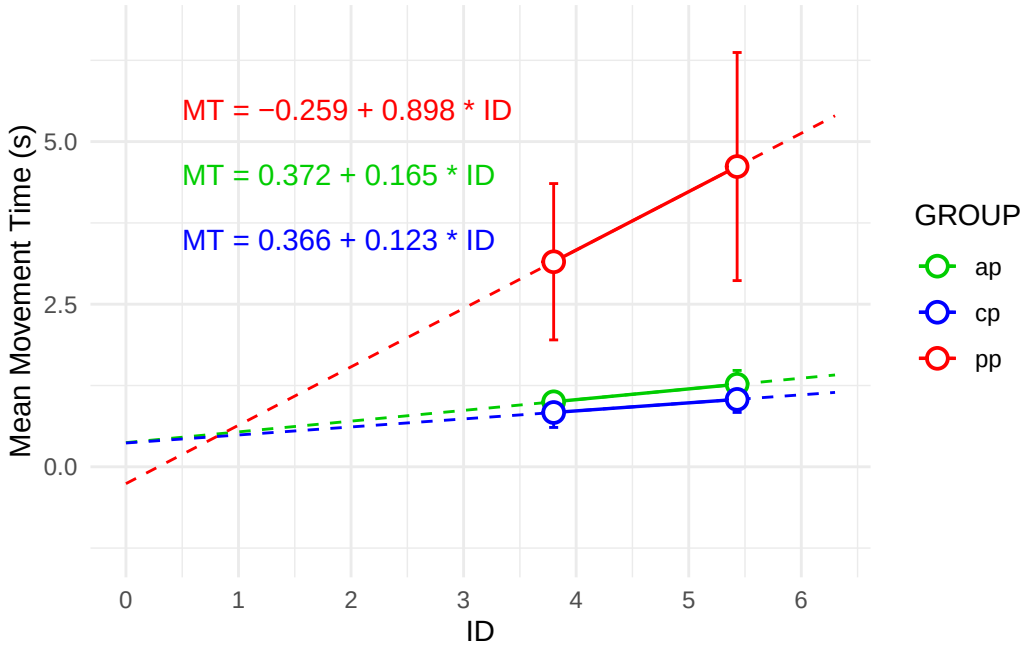
  # Axes and legend
  scale_x_continuous(name = "ID", breaks = 0:6, limits = c(0, 6.3)) +
  scale_y_continuous(name = "Mean Movement Time (s)", limits = c(-1.3, 6.7)) +

  # Group colors
  scale_color_manual(values = c("pp" = "red", "ap" = "green3", "cp" = "blue")) +

  labs(color = "GROUP") +
  theme_minimal()

show(publication_plot)
```

```
`geom_smooth()` using formula = 'y ~ x'
```



The **solid lines** connect the measured mean values for `ID1` and `ID2`.

The **dashed lines** show the linear regression for each group.

The error bars show:

$Mean \pm 0.5 \times SD$

Larger error bars mean that the Movement Time values are more spread out around the mean.

1.6.1 Interpretation

Movement Time increases with ID in all three groups, but the increase is much larger for the stroke group (`pp`). The `ap` and `cp` groups show smaller and relatively similar increases.

The `pp` group also shows much greater variability in Movement Time, as indicated by its larger error bars, whereas the healthy groups have smaller error bars and more consistent values.

Overall, the figure visually supports the regression results and the `GROUP × ID` interaction described previously.

1.7 Discuss the differences with the article

1.7.1 Comparison with the article

Our analysis aimed to reproduce the main findings of Mottet et al. (2017), who studied movement time in healthy young adults, healthy age-matched adults, and participants with stroke.

Our mixed-design ANOVA showed a significant effect of GROUP, $F(2, 48) = 37.24$, $p < 0.001$, and ID, $F(1, 48) = 26.11$, $p < 0.001$. The GROUP × ID interaction was also significant, $F(2, 48) = 11.58$, $p < 0.001$. These results indicate that movement time differed between groups, increased with difficulty, and that the increase was not the same across all three groups.

The regression analysis showed positive slopes in all three groups, with the steepest slope in the stroke group. The detailed comparison with the original article is presented below.

The original article by Mottet et al. (2017) also found a significant effect of GROUP on movement time, $F(2, 48) = 35.32$, $p < 0.001$, and a significant effect of ID, $F(1, 48) = 5.42$, $p < 0.001$. The GROUP × ID interaction was also significant, $F(2, 48) = 10.02$, $p < 0.001$. The authors reported Fitts's law slopes of 0.118 for age-matched controls, 0.097 for young controls, and 0.672 for participants with stroke.

Our regression slopes were 0.165 for healthy adults (ap), 0.123 for healthy young adults (cp), and 0.898 for the stroke group (pp). In both analyses, the stroke group had the steepest slope, showing a stronger increase in movement time with difficulty. However, our slope estimates were higher than those reported in the article. This difference may reflect the simpler analysis

and reduced dataset used in our assignment, so the numerical values should not be interpreted as directly equivalent. Despite these differences, both analyses show the same overall pattern: movement time increases with difficulty, and this relationship is stronger in participants with stroke.

However, our analysis is simpler than the original study. We retained only ID1 and ID2 because ID3 was incomplete for the stroke group, and we removed R3 because it was not completed consistently across all groups. We then averaged movement time for each participant and difficulty condition. The original article considered additional task factors, including movement direction and orientation, and used post-hoc comparisons to examine differences between groups.

Therefore, our results support a similar general pattern, but the numerical results cannot be compared directly because the analyses differ. In addition, we did not perform post-hoc pairwise comparisons, so we cannot determine from our ANOVA alone which specific pairs of groups differ significantly.

1.7.2 Conclusion

Our analysis showed that movement time increased with task difficulty and differed between the three groups. The significant GROUP $\times$ ID interaction indicates that the effect of difficulty was not the same across all groups. Both the ANOVA and the regression graph showed a larger increase in movement time in the stroke group.

These findings are consistent with the main conclusion of Mottet et al. (2017). Although our analysis used a reduced dataset and a simpler approach, it showed the same overall pattern as the original article. The regression slopes were higher in our analysis, especially in the stroke group, but the overall relationship between movement time and task difficulty was similar. Further pairwise comparisons would be needed to determine exactly which groups differ significantly from one another.

Overall, this assignment helped us understand how to clean and reorganize movement data, perform a mixed-design ANOVA, calculate group-specific linear regressions, visualize the results, and compare our findings with published research.

2 Git Workflow Strategy

3 Challenges and Lessons Learned

Once more the challenges was to cross our different code by checking the variable names and the data structure. We also had to check the data for missing values and outliers, which was a bit tricky. But we learned a lot about data cleaning and manipulation, as well as how to perform

ANOVA analysis in R. We also learned how to interpret the results and create visualizations to present our findings.

4 Data Sources

5 Scientific References

6 Grading Checklist